

# Yijie Xu 徐亦捷

Ph.D. Candidate in Artificial Intelligence at HKUST(GZ)

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## RESEARCH INTEREST

His current research interests are **Test-time/Self-Evolving Language Models** and **Retrieval-Augmented Generation (RAG)**; he also studies how humans and AI collaborate and interact.

## EDUCATION

**Hong Kong University of Science and Technology (Guangzhou)**

Guangzhou, China

PH.D. CANDIDATE IN ARTIFICIAL INTELLIGENCE

Aug. 2027 Expected

- **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.
- **Co-supervisor:** Prof. Xuming Hu, Assistant Professor in AI Thrust, HKUST(GZ).

**Hong Kong University of Science and Technology (Guangzhou)**

Guangzhou, China

M.PHIL. IN ARTIFICIAL INTELLIGENCE

Sep. 2022 - Aug. 2024

• **CGPA:** 3.96/4.3

• **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.

• **Co-supervisor:** Prof. Xiaojuan Ma, Associate Professor in CSE, HKUST.

**University of Reading &**

Reading, England / Nanjing, China

**Nanjing University of Information Science and Technology**

DUAL B.SC. IN MATHEMATICS AND APPLIED MATHEMATICS

Sep. 2018 - Jun. 2022

• **Dual-degree** program.

• **CGPA:** 3.71/4.

• **Final Year Project:** Research on Interpolation and Data Fitting: Basis and Applications.

## PUBLICATIONS

†: Co-first Authors; \*: Corresponding Authors.

- [1] **You only need 4 extra tokens: Synergistic Test-time Adaptation for LLMs**  
Yijie Xu, Huizai Yao, Zhiyu Guo, Weiyu Guo, Pengteng Li, Aiwei Liu, Xuming Hu, Hui Xiong\*  
Under Review for ICML'26  
Honored to be cited by Prof. Yann LeCun and Jitendra Malik as a representative exploration of LLM test-time adaptation.
- [2] **SePer: Measure Retrieval Utility Through The Lens Of Semantic Perplexity Reduction**  
Lu Dai, Yijie Xu, Jinhui Ye, Hao Liu\*, Hui Xiong\*  
ICLR'25 **Spotlight(3.20%)**: The Thirteenth International Conference on Learning Representations
- [3] **Gated Test-Time Scaling for GUI Grounding via Multi-Seed Zoom Search**  
Huizai Yao, Yijie Xu, Tianfu Wang, Yili Wang, Yi Cui, Yunfan Lu, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun\*, Hui Xiong\*  
Under Review for ICML'26
- [4] **Mark Your LLM: Detecting the Misuse of Open-Source Large Language Models via Watermarking**  
Yijie Xu†, Aiwei Liu†, Xuming Hu\*, Lijie Wen, Hui Xiong\*  
WMARK@ICLR2025: The 1st Workshop on GenAI Watermarking, ICLR 2025
- [5] **Against Political Polarization: A Unified Framework for Tracing Evolving Political Ideologies on Social Media**  
Yijie Xu, Chao Wang\*, Hui Xiong\*  
TIST: ACM Transactions on Intelligent Systems and Technology. (Accepted with Minor Revision)
- [6] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining**  
Yi Cui†, Zilin Wang†, Yijie Xu, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhuo Zhong\*, Hui Xiong\*  
Under Review for KDD Datasets & Benchmarks Track'26
- [7] **Revisiting Noise Resilience Strategies in Gesture Recognition: Short-Term Enhancement in sEMG Analysis**  
Weiyu Guo, Ziyue Qiao, Ying Sun\*, Yijie Xu, Hui Xiong\*  
ICML'25: The Forty-Second International Conference on Machine Learning
- [8] **Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding**  
Weiyu Guo, Ziyang Chen, Shaoguang Wang, Jianxiang He, Yijie Xu, Jinhui Ye, Ying Sun\*, Hui Xiong\*  
NeurIPS'25: The Thirty-ninth Annual Conference on Neural Information Processing Systems
- [9] **See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model**  
Pengteng Li, Pinhao Song, Wuyang Li, Huizai Yao, Weiyu Guo, Yijie Xu, Dugang Liu, Hui Xiong\*  
NeurIPS'25: The Thirty-ninth Annual Conference on Neural Information Processing Systems
- [10] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection**  
Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong\*  
AAAI'26: The Fortieth AAAI Conference on Artificial Intelligence

[11] **Less is More: Token-Efficient Video-QA via Adaptive Frame-Pruning and Semantic Graph Integration**  
Shaoguang Wang, Jianxiang He, Weiyu Guo, Ziyang Chen, Yijie Xu, Xuming Hu, Hui Xiong\*  
*Findings of CVPR’26: The IEEE / CVF Computer Vision and Pattern Recognition Conference*

[12] **SpGesture: Source-Free Domain-adaptive sEMG-based Gesture Recognition with Jaccard Attentive Spiking Neural Network**  
Weiyu Guo, Ying Sun\*, Yijie Xu, Ziyue Qiao, Yongkui Yang, Hui Xiong\*  
*NeurIPS’24: The Thirty-eighth Annual Conference on Neural Information Processing Systems*

[13] **Event Camera Demosaicing via Swin Transformer and Pixel-focus Loss**  
Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong\*  
*CVPRW’24: Workshop for IEEE / CVF Conference on Computer Vision and Pattern Recognition*

[14] **MIPI 2024 Challenge on Demosaic for HybridEVS Camera: Methods and Results**  
Yaqi Wu, et al., Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong, et al.  
*CVPRW’24: Workshop for IEEE / CVF Conference on Computer Vision and Pattern Recognition*

[15] **Contrastive Graph Matching for Domain Adaptive Object Detection**  
Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, Yijie Xu, Minghao Yang, Pinhao Song  
*arXiv*

SELECTED PROJECTS

- Construct Supervision: Regulation-Grounded LLM System**

Sep. 2025 - Present / Poly Group

  - **Screened** and structured evidence from 3,000+ legal and regulatory documents for construction supervision.
  - **Built** regulation-grounded retrieval and assessment workflows and frontend web interface for downstream RAG and evaluation.
- Multimodal HCI Perception Modeling**

Jan. 2023 - Aug. 2024 / MetaHCI

  - **Led** platform setup, data collection, algorithm design, and group coordination for an RBM multimodal HCI project.
  - **Developed** LLM-for-AIGC workflows, including training, visualization, deployment, and service maintenance for MetaHCI, a HKUST(GZ)-affiliated startup valued at 70 million RMB.
- Political Ideology Analysis on Social Media**

Nov. 2022 - Oct. 2023

  - **Built** a 230k+ report dataset from millions of tweets and author profiles.
  - **Proposed** a NLP + Temporal Graph Neural Network framework for ideology detection and prediction.
- Big Data Mining Services for CRM Systems**

Nov. 2022 - May. 2023 / Tencent

  - **Built** web-crawling and pre-trained LM pipelines for large-scale academic talent profiling.
  - **Operationalized** extracted profiles for an industry-facing CRM project at Tencent.

SELECTED SCHOLARSHIPS AND AWARDS

- Research Travel Grant

Apr. 2025
- Doctoral Scholarship Award

Sep. 2024
- Postgraduate Scholarship Award

Sep. 2022
- First Prize of the Smart Audiovisual Track, National AI Application Innovation Challenge

Sep. 2024
- Champion of the HCI channel, the 2nd Guangzhou Pazhou Algorithm Competition (Top 15/3307)

Sep. 2023
- Scholarship from Nanjing University of Information Science and Technology

Nov. 2021
- Scholarship from China Mobile Communications Corporation (CMCC)

Nov. 2020
- Scholarship from China Mobile Communications Corporation (CMCC)

Nov. 2019

TEACHING EXPERIENCE

- HKUST(GZ)**

Big Data Applications for Business and Government (Spring 2025)      Natural Language Processing and Its Applications (Fall 2024)

Red Bird Challenge Camp (Summer 2022, 2023)
- NUIST & Reading**

Professional Skills for Mathematicians (Fall 2021)      Numerical Analysis (Fall 2021)

Applied Stochastic Process (Fall 2021)      Mathematical Programming (Spring 2021)

SERVICES & SKILLS

- PC

2026, COLM, ICML, CVPR, ICLR, AAAI, ACM MM

2025, WWW, AAAI, ACM MM, KDD, NeurIPS, ICML, IJCAI, ICLR, ICASSP, AISTATS, ICWSM

2024, NeurIPS
- Review

ACM Computing Surveys; IEEE Transactions on Circuits and Systems for Video Technology
- Campus

Student Representative, Committee on Undergraduate Studies, HKUST(GZ) Senate (2024–2025)
- Stack

Python; PyTorch; Transformers; LLaMA Factory; vLLM & SGLang; LLM MCP&Agents; Web Crawling
- Languages

Chinese; English (IELTS 7)